│ ├── services/ # Shared services │ │ ├── llm/ │ │ ├── memory/ │ │ ├── vector/ │ │ ├── scrapers/ │ │ └── integrations/ │ │ │ ├── shared/ # Shared utilities │ │ ├── types/ │ │ ├── constants/ │ │ ├── utils/ │ │ └── errors/ │ │ │ └── workers/ # Background jobs │ ├── queue.ts │ └── processors/ │ ├── config/ # Configuration │ ├── database.ts │ ├── redis.ts │ ├── llm.ts │ └── agents.ts │ ├── docs/ # Documentation │ ├── api.md │ ├── agents.md │ ├── integrations.md │ └── deployment.md │ ├── tests/ # Test suites │ ├── unit/ │ ├── integration/ │ └── e2e/ │ ├── scripts/ # Utility scripts │ ├── migrate.sh │ ├── seed.sh │ └── deploy.sh │ ├── docker-compose.yml ├── Dockerfile ├── package.json ├── tsconfig.json └── README.md ```

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7. Key Design Decisions & Trade-offs

7.1 Why Node.js/NestJS over Python?

Node.js chosen because: - Single language across stack (TypeScript everywhere) - Better async/concurrent request handling for I/O-bound workloads - Native GraphQL support via Apollo - Large ecosystem for real-time applications

Trade-off: Python has better ML libraries, but we can call Python microservices for specific ML tasks if needed.

7.2 Why GraphQL over REST?

GraphQL chosen because: - Mobile and web clients need different data shapes - Real-time subscriptions built-in - Strong typing reduces bugs - Single endpoint simplifies architecture

Trade-off: Caching is more complex; we'll use persisted queries and DataLoader pattern.

7.3 Multi-tenancy Approach

Hybrid strategy: - Start with row-level security (simplest, cost-effective) - Migrate enterprise clients to schema-per-tenant as needed - This balances cost and isolation requirements

7.4 Agent Coordination

Event-driven vs Direct API calls: - Event-driven chosen for loose coupling and scalability - Each agent can be deployed independently - Easier to add new agents without modifying existing ones

7.5 LLM Strategy

Multi-provider approach: - GPT-4o for complex reasoning tasks - Claude for long-form content generation - GPT-4o-mini for high-volume, simple tasks - This optimizes cost/quality balance

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8. Security & Compliance

8.1 NZ-Specific Requirements

- **Privacy Act 2020**: Data retention limits, access controls
- **REAA 2008**: Agent licensing verification, audit trails
- **Anti-money laundering**: Customer due diligence workflows

8.2 Technical Measures

- End-to-end encryption for sensitive data
- Field-level encryption for PII
- Audit logging for all agent decisions
- Regular penetration testing

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9. Scaling Strategy

Phase 1 (0-500 agents)

- Single deployment, managed PostgreSQL/Redis
- All agents in one process
- Cost: ~$500/month infrastructure

Phase 2 (500-5,000 agents)

- Agent workers scale independently
- Read replicas for database
- CDN for static assets
- Cost: ~$2,000/month

Phase 3 (5,000+ agents)

- Schema-per-tenant for enterprise
- Regional deployments (AU/NZ)
- Dedicated ML inference cluster
- Cost: ~$10,000+/month

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10. Development Roadmap

Sprint 1-2: Foundation

- [ ] Database schema + migrations
- [ ] API skeleton (GraphQL)
- [ ] Auth integration (Clerk)
- [ ] Base agent framework

Sprint 3-4: Lead Agent MVP

- [ ] Intent classification
- [ ] Lead qualification flow
- [ ] CRM connector (AgentBox)
- [ ] WhatsApp integration

Sprint 5-6: Content Agent MVP

- [ ] Property description generator
- [ ] SEO optimization
- [ ] Image analysis
- [ ] Social post templates

Sprint 7-8: Scheduling & Transaction

- [ ] Calendar integration
- [ ] Inspection booking
- [ ] Transaction pipeline
- [ ] Document templates

Sprint 9-10: Market Intelligence

- [ ] TradeMe scraper
- [ ] CMA generation
- [ ] Price alerts
- [ ] Market reports

Appendix: NZ Real Estate Context

Key Terms

- **LIM**: Land Information Memorandum (council records)
- **S&P**: Sale and Purchase Agreement
- **REINZ**: Real Estate Institute of New Zealand
- **REAA**: Real Estate Agents Authority
- **Open Home**: Scheduled property viewing
- **Tender**: Closed bidding process
- **Auction**: Public bidding process (very common in NZ)
- **Settlement**: Closing/completion date
- **Conditions**: Finance, building report, LIM, valuation

Regional Variations

- Auckland: Fastest market, auctions common
- Wellington: Government influence, steady market
- Christchurch: Post-quake rebuild focus
- Regional NZ: Longer sales cycles, more conditional offers